MAASAI MARA UNIVERSITY

REGULAR UNIVERSITY EXAMINATIONS
2023/2024 ACADEMIC YEAR
SECOND YEAR FIRST SEMESTER

SCHOOL OF PURE APPLIED AND HEALTH SCIENCES
BSC COMPUTER SCIENCES

COURSE CODE: COM 2110-1

COURSE TITLE: OBJECT ORIENTED PROGRAMMING 1

DATE: 15TH DEC 2023 **TIME: 8:30AM-10:30AM**

INSTRUCTIONS TO CANDIDATES

Answer Question ONE and any other TWO questions

QUESTION 1 (20 MARKS)

a) Discuss citing examples the following concepts as used in C++: (2mks)
   i. Function overriding
   ii. Inheritance

b) Write a code to declare a function that calculates the area of a rectangle and discuss the parts that make up the function. (2mks)

c) Explain the purpose or use of the return statement in a function. (2mks)

a) Discuss THREE access modifiers used in C++ Programming. (3mks)

b) What is an integer expression ? (2mks)

c) Define the term array, discuss how array elements are accessed and write a statement that declares and initializes an array to store the marks of five students. (4mks)

d) What will be the output of the following program? (1mks)

```
#include <iostream>
using namespace std;
int main()
{
  int x;  // variable declaration.
   int y;  // variable declaration
    int z;  // variable declaration
   cout<<"Enter the values of x and y";
   cin>>x>>y;
   z=x+y;
    cout<<"\n"<<"Value of z is :"<<z; //  displaying the value of z.
    return 0;
}
```

Explain your answer (2mks)

e) C++ is a statically typed, compiled, general-purpose, case-sensitive, free-form programming language that supports **procedural**, **object-oriented**, **and generic programming.** Differentiate between procedural, object oriented and generic programming (2 Marks)

QUESTION TWO **(15 MARKS)**

a) Write a C++ code When the code is compiled and executed, it produces results as follows (3 Marks)

```
Values of 'a' and 'b' are : 80,80
```

b) C++ fully supports object-oriented programming, including the following four pillars of object-oriented development briefly discuss the following four pillars (3 marks)

i. Encapsulation

ii. Data hiding

iii. Inheritance

iv. Polymorphism

c) Differentiate the following C++ parts (3 marks)

i. **Object**
ii. **Class**
iii. **Methods**
iv. **Instance Variables**

d) what is a decision making statement give at least two examples (3 marks)

e) Differentiate the following pointers as used in C++ (3 marks)

i. Null Pointers
ii. Pointer Arithmetic
iii. Pointers vs Arrays
iv. Array of Pointers

f)

QUESTION THREE (15 MARKS)

g) Explain the following various parts of C++ program? (3 Marks)

i. **<iostream>**
ii. **'// main**
iii. **int main**
iv. **return 0;**

h) Discuss any three reserved key words in C++. (3 Marks)

i) Whats a trigraph sequence? (3 Marks)

j) C++ offers the programmer a rich assortment of built-in as well as user defined data types. Following table lists down seven basic C++ data types –

k) Write a C++ programme where a variable has been declared at the top, but it can been defined inside the main function (6 Marks)

QUESTION FOUR 15 MARKS

l) Whats Variable Declaration in C++? (3marks)

m) Explain the following code into details and the output when code is compiled and executed, (3 marks)

```
#include <iostream>
using namespace std;

#define LENGTH 10
#define WIDTH  5
#define NEWLINE '\n'

int main() {
  int area;

  area = LENGTH * WIDTH;
  cout << area;
  cout << NEWLINE;
  return 0;
}
```

n) C++ is rich in built-in operators differentiate the following operator as used in C++ (3marks)

i. Arithmetic Operators
ii. Relational Operators
iii. Logical Operators
iv. Bitwise Operators
v. Assignment Operators
vi. Misc Operators

o) Differentiate the following loops to handle looping requirements. (3 Marks)

i. for loop
ii. do...while loop
iii. nested loops
iv. while loops

p) What's and array? and how do you Declare an in C++, (3 Marks)